

A Comprehensive Risk Engine for VaR and ES Calculations on Equity and Option Portfolios

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Abstract

This paper presents a portfolio risk engine for estimating Value at Risk (VaR) and Expected Shortfall (ES) for portfolios containing both equity and European option positions. The framework combines historical market data, instrument-level valuation, and portfolio aggregation in a single implementation, with stocks marked from observed price histories and options repriced using the Black-Scholes model with underlying prices, implied volatility inputs, risk-free rates, and time to expiry. By explicitly incorporating derivative positions, the engine captures nonlinear features that materially affect tail-risk measurement in mixed portfolios. The project serves both as a practical computational framework and as a model documentation study, formalizing the portfolio, pricing assumptions, and risk measurement structure used to evaluate downside risk in a stock-option setting.

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1 Introduction

This project implements a portfolio risk engine for estimating the Value at Risk (VaR) and Expected Shortfall (ES) for portfolios containing both equity and European option positions. The codebase is built around a full mark-to-market workflow: stock positions are valued from historical price data, option positions are repriced using Black-Scholes, and portfolio losses are aggregated over a chosen holding period and confidence level. Rather than treating risk as a purely statistical quantity, the project ties risk measurement directly to instrument-level valuation.

A key feature of the engine is that options are handled explicitly rather than folded into a simplified stock-only approximation. The code tracks each option's underlying, strike, expiry, implied volatility input, and risk-free rate, so portfolio risk reflects nonlinearity, time-to-maturity effects, and changing market inputs, which makes this similar to a small research-oriented risk system than a basic VaR calculator.

The scope is intentionally focused. The engine works with daily market data, listed equities, European-style options, Treasury-yield-based risk-free rates, and volatility surface inputs stored locally in the project data layer. On top of that foundation, it supports historical, parametric and Monte Carlo methods of constructing the portfolio loss distribution with different VaR and ES methodologies.

The purpose of this paper is to document that framework clearly: how the portfolio is represented, how instruments are repriced, what assumptions enter the market and model layers, and how those choices affect the final tail-risk estimates. The emphasis is therefore not only on defining VaR and ES, but on explaining how this particular implementation computes them in a portfolio setting with both linear and nonlinear exposures.

2 Problem Setup

Our objective is to estimate the downside risk of a portfolio containing equity and European option positions over a fixed holding period. Let t denote the valuation date and let h denote the risk horizon measured in trading days. The portfolio is marked to market on date t , then revalued under a distribution of future market scenarios over the interval $[t, t + h]$ to get the resulting profit-and-loss (PnL).

The two primary risk measures studied are Value at Risk (VaR) and Expected Shortfall (ES). For a confidence level $\alpha \in (0, 1)$, the VaR at level α is the loss threshold such that only a fraction $1 - \alpha$ of the loss distribution lies beyond it. Expected Shortfall measures the average loss conditional on losses exceeding the VaR threshold. Both quantities are computed from the portfolio loss distribution rather than from individual instruments in isolation.

The portfolio can contain long and short positions in stocks and European options. Risk is evaluated under a historical lookback window (determines how much historical information is used for calibration) and a forecast horizon (determines the period over which losses are measured). In the code, these are `window_size` and `up_to_days` respectively.

Problem Statement: Given a portfolio observed on date t , together with historical market data and model inputs, construct a distribution for the portfolio PnL over horizon h , and from that distribution estimate VaR and ES at confidence level α .

3 Portfolio Valuation Framework

Risk measurement in this project is based on full mark-to-market portfolio valuation. On each valuation date t , the total portfolio value is computed as the sum of the values of all stock and option positions. Linear instruments are valued directly from observed market prices, while nonlinear option positions are repriced using the Black-Scholes model. Suppose the mark-to-market portfolio value at time t is V_t .

Let the portfolio contain N_s stock positions and N_o option positions.

$$V_t = \sum_{i=1}^{N_s} q_i^{(s)} S_{i,t} + \sum_{j=1}^{N_o} q_j^{(o)} O_{j,t}, \quad (1)$$

where $q_i^{(s)}$ denotes the signed quantity of stock position i , $S_{i,t}$ denotes its market price at time t , $q_j^{(o)}$ denotes the signed quantity of option position j , and $O_{j,t}$ denotes the model value of that option at time t .

For stock positions, valuation is direct since we obtain stock prices from stored historical close data. If a stock with ticker i has market price $S_{i,t}$ on date t , then the contribution of that position is

$$V_{i,t}^{(s)} = q_i^{(s)} S_{i,t}. \quad (2)$$

For option positions, we use Black-Scholes repricing. If option j has underlying price S_t , strike K , time to expiry T_t , risk-free rate r_t , implied volatility σ_t , option type $\tau \in \{\text{call}, \text{put}\}$, the value is

$$O_{j,t} = BS(S_t, K, T_t, r_t, \sigma_t, \tau). \quad (3)$$

For a European call option, Black-Scholes with $\Phi(\cdot)$ denotes the standard normal c.d.f gives

$$C_t = S_t \Phi(d_1) - K e^{-r_t T_t} \Phi(d_2), \quad P_t = K e^{-r_t T_t} \Phi(-d_2) - S_t \Phi(-d_1) \quad (4)$$

$$d_1 = \frac{\ln(S_t/K) + (r_t + \frac{1}{2}\sigma_t^2) T_t}{\sigma_t \sqrt{T_t}}, \quad d_2 = d_1 - \sigma_t \sqrt{T_t}. \quad (5)$$

This pricing step is central to the project because it allows the risk engine to capture the nonlinear dependence of option value on the underlying, volatility, and time to maturity.

The time-to-expiry input is measured in trading-year units using the convention $T_t = \frac{\text{days until expiry}}{252}$. This 252-day convention is used throughout for valuation, scenario generation, and backtesting.

If an option is at or beyond expiry, then the Black-Scholes formula is no longer applied and the option is instead valued at intrinsic value. For a call and put, $C_t = \max(S_t - K, 0)$ and $P_t = \max(K - S_t, 0)$.

Given the mark-to-market portfolio value at the valuation date t and the revalued portfolio at horizon $t + h$, the portfolio profit-and-loss over the horizon and portfolio loss is

$$\text{PnL}_{t,t+h} = V_{t+h} - V_t, \quad L_{t,t+h} = -\text{PnL}_{t,t+h} = -(V_{t+h} - V_t) \quad (6)$$

The full project is therefore organized around constructing a distribution for $\text{PnL}_{t,t+h}$, or equivalently $L_{t,t+h}$, under different scenario-generation methodologies. VaR and ES are then computed from that portfolio-level loss distribution rather than from isolated asset returns. This valuation framework is what allows the engine to treat stocks and options within a single coherent risk system.

4 Data and Market Inputs

The risk engine is built on a daily market-data framework. Historical equity prices are used to value stock positions, construct return and shock series, and calibrate the historical, Monte Carlo, and parametric risk models. For each underlying asset, we store daily close data as the fundamental price input, and all scenario generation is aligned to the common set of available trading dates across the portfolio universe.

Risk-free interest rates are incorporated through a daily Treasury-yield proxy, specifically a 3-month Treasury series stored locally in the project data layer. These rates enter the Black-Scholes valuation of option positions and are also used in scenario-based risk calculations when rate shocks are included.

Option market inputs are handled through locally stored daily option surface files containing contract-level information such as strike, expiry, option type, and observed close price. Because implied volatilities are not always directly available as raw inputs, the project infers them from option prices and then constructs interpolated volatility surfaces for valuation and historical shock analysis.

5 Implied Volatility Surface Construction

A central difficulty in valuing option portfolios is that market option data are naturally observed in price space, whereas model-based repricing requires volatility inputs. For this reason, the project constructs implied volatility surfaces from daily option price data. On each surface date t , the raw option dataset contains contract-level information such as strike K , expiry date T^{exp} , option type, and observed market close price. The goal is to convert these prices into implied volatilities and then organize them into interpolated surfaces that can be queried during risk calculations.

5.1 Implied Volatility Inversion Methodology

For a given option contract, implied volatility is defined as the value of σ that makes the Black-Scholes model price equal to the observed market price. If P_t^{mkt} denotes the observed option price and $BS(S_t, K, \tau_t, r_t, \sigma, \text{type})$ denotes the Black-Scholes price, then the implied volatility σ_t^{imp} solves

$$BS(S_t, K, \tau_t, r_t, \sigma_t^{\text{imp}}, \text{type}) - P_t^{\text{mkt}} = 0, \quad (7)$$

where S_t is the underlying spot price, r_t is the risk-free rate, and τ_t is the time to expiry in years.

Because this equation does not admit a closed-form solution for σ , the project solves it numerically for each option contract in the daily price surface using a hybrid root-finding procedure. First, Newton's method is applied because it is fast when the initial guess is reasonable. Given an iterate σ_n ,

$$\sigma_{n+1} = \sigma_n - \frac{f(\sigma_n)}{f'(\sigma_n)}, \quad (8)$$

where $f'(\sigma_n)$ is derivative of the option pricing error w.r.t. volatility, i.e. the option vega,

$$f'(\sigma) = \frac{\partial BS}{\partial \sigma}. \quad (9)$$

If Newton's method fails to converge, the algorithm falls back to Brent's method, which is slower but more numerically robust. This two-stage procedure balances computational efficiency with numerical stability when processing large daily option surfaces.

Not every raw option quote is suitable for implied-volatility inversion. The project therefore applies a validity filter before solving for σ . In particular, contracts are excluded if time to maturity is non-positive, if market prices are missing, or if the observed price is below intrinsic value up to numerical tolerance. This avoids attempting to invert economically invalid or numerically unstable quotes.

5.2 Valuation Surface

Once implied volatilities have been computed for valid contracts, the project constructs an interpolated surface for option valuation. This surface is represented in strike-maturity coordinates,

$$\sigma = \sigma(K, \tau), \quad (10)$$

where τ is the remaining time to expiry in years. This is the natural coordinate system for Black-Scholes repricing, since option valuation depends directly on remaining maturity rather than on calendar expiry alone.

Interpolation is performed over the set of observed $(K, \tau, \sigma^{\text{imp}})$ points using a multidimensional interpolator. If the requested maturity lies outside the observed maturity range, linear extrapolation is applied in the maturity direction using the nearest boundary slices. This allows pricing layer to query implied volatility even when the exact maturity is not directly represented in the observed daily surface.

5.3 Fixed-Expiry Surface for Historical Volatility Shocks

A second surface representation is used for historical volatility-shock construction. In that case, the relevant object is not a moving maturity surface but a fixed-expiry surface,

$$\sigma = \sigma(K, T^{\text{exp}}), \quad (11)$$

where the calendar expiry date is held fixed.

For option valuation, implied volatility is represented as a function of strike and remaining time to maturity because this is the coordinate system required by Black-Scholes repricing. For historical volatility shock construction, however, the project instead uses a fixed-expiry surface, in which implied volatility is indexed by strike and calendar expiry date. This distinction prevents historical volatility shocks from being mechanically distorted by maturity roll-down as time passes. In this way, the code uses one surface representation for valuation and a second, related representation for extracting cleaner historical implied-volatility shocks.

5.4 Computational Considerations

This part of the code is among the most computationally expensive in the project. We must repeatedly infer implied volatilities from historical option prices, construct multiple interpolators, and query those interpolators across many contracts and dates. Much of this complexity arises because historical implied-volatility surface data are difficult to obtain directly without access to expensive commercial datasets. As a result, the project reconstructs the required volatility inputs by starting from historical option prices and reverse-engineering implied volatilities before building the relevant surface representations.

6 Historical Simulation Methodology

Historical simulation estimates the portfolio loss distribution by reusing realized market moves from the historical record rather than imposing a fully parametric law for returns. In contrast to a purely statistical historical VaR built only from portfolio return series, we apply historical shocks at the market-input level and revalue current portfolio under those shocked conditions. This is essential for option positions where risk depends not only on underlying price change but also implied vol and interest-rate movements.

Let t denote the forecast date, h denote the risk horizon in trading days, and W denote the historical lookback window. We want to construct a sequence of scenario PnLs

$$\left\{ \Pi_t^{(1)}, \Pi_t^{(2)}, \dots, \Pi_t^{(W)} \right\}, \quad (12)$$

where each $\Pi_t^{(k)}$ is the hypothetical portfolio profit-and-loss that would result if the current portfolio, observed on date t , were subjected to a market shock analogous to the historical movement observed over one past window of length h . VaR and ES are then computed from this empirical scenario distribution.

6.1 Historical Scenario Construction

Historical simulation begins by selecting a sequence of past scenario windows from the aligned market-data history. For a forecast date t , the code first loads historical daily data for all stock tickers and option underlyings that are relevant to the portfolio. These time series are aligned on a common set of dates so that each scenario uses a consistent cross-section of market inputs. Let $\{t_1, t_2, \dots, t_{W+h}\}$ denote the aligned historical dates available up to the forecast date, ordered chronologically. A horizon- h historical shock is then built from pairs of dates (t_k, t_{k+h}) for $k = 1, \dots, W$.

For stock positions, the historical scenario is based on observed price changes in the underlying equity. If $S_{u,\tau}$ denotes the price of underlying u on date τ , then the stock shock in scenario k is

$$\Delta S_u^{(k)} = S_{u,t_{k+h}} - S_{u,t_k}. \quad (13)$$

This shock is then applied to the current portfolio state rather than to the historical portfolio itself. In that sense, the methodology is a current-portfolio historical shock simulation rather than a simple replay of historical portfolio values. Risk-free rates are treated similarly. Let r_τ denote the observed short-term risk-free rate on date τ . The historical rate shock in scenario k is

$$\Delta r^{(k)} = r_{t_{k+h}} - r_{t_k}. \quad (14)$$

These rate shocks are used directly in option repricing so that the historical scenarios incorporate both price and discount-rate variation.

6.2 Historical Implied Volatility Shocks

For option portfolios, the most delicate part of historical simulation is the treatment of implied volatility. A naive approach would compute volatility changes using the standard strike-maturity surface on each historical date. However, this would confound true volatility movement with the mechanical roll-down in time to maturity that occurs as calendar time advances. To avoid this issue, the project extracts historical implied-volatility shocks from a fixed-expiry surface.

Let $\sigma_t(K, T^{\text{exp}}, \tau)$ denote the implied volatility observed on date t for strike K , calendar expiry T^{exp} , and option type $\tau \in \{\text{call}, \text{put}\}$. In the historical shock engine, the expiry date is held fixed and the volatility shock for scenario k is defined as

$$\Delta\sigma^{(k)} = \sigma_{t_{k+h}}(K, T^{\text{exp}}, \tau) - \sigma_{t_k}(K, T^{\text{exp}}, \tau). \quad (15)$$

This construction measures how the market repriced the same expiry bucket across two historical dates, rather than how volatility changes when one moves mechanically down the maturity axis. As a result, the historical volatility shock more cleanly reflects changes in market-implied risk.

In implementation terms, daily implied-volatility surfaces are first inferred from option prices, then queried through a fixed-expiry interpolator. The historical shock computation therefore uses the same daily surface data as valuation, but a different interpolation coordinate system. This is a deliberate methodological choice intended to separate historical volatility movement from passage-of-time effects.

6.3 Current-Portfolio Shock Revaluation

Once the historical shocks are constructed, they are applied to the current portfolio state observed on date t . For a stock position in underlying u with current spot price $S_{u,t}$, the shocked underlying price in scenario k is

$$S_{u,t}^{(k)} = S_{u,t} + \Delta S_u^{(k)}. \quad (16)$$

Likewise, the shocked risk-free rate is

$$r_t^{(k)} = r_t + \Delta r^{(k)}, \quad (17)$$

and for an option contract with current implied volatility σ_t , the shocked implied volatility is

$$\sigma_t^{(k)} = \sigma_t + \Delta\sigma^{(k)}. \quad (18)$$

The project therefore treats historical simulation as a factor-shock methodology: observed past changes in market inputs are mapped onto the current portfolio configuration, and the portfolio is then fully revalued under those shocked inputs.

A further modeling choice concerns time to expiry. In the current implementation, the time-to-maturity input used for option repricing is held fixed at its value on the forecast date. If an option expires on calendar date T^{exp} , then the valuation maturity input is

$$\tau_t = \frac{T^{\text{exp}} - t}{252}, \quad (19)$$

and this same τ_t is used across all historical scenarios. Thus the historical engine shocks the underlying price, risk-free rate, and implied volatility while keeping the maturity coordinate fixed. This means the historical method is best interpreted as a current-state factor-shock revaluation rather than as a full forward horizon repricing with theta decay.

For an option position with strike K , current maturity τ_t , option type τ , and quantity q , the scenario- k shocked option value is

$$O_t^{(k)} = BS\left(S_t^{(k)}, K, \tau_t, r_t^{(k)}, \sigma_t^{(k)}, \tau\right). \quad (20)$$

The option contribution to scenario PnL is then

$$\Pi_{t,\text{option}}^{(k)} = q\left(O_t^{(k)} - O_t\right), \quad (21)$$

where O_t is the current mark-to-market value of the option. For stock positions, the scenario PnL takes the simpler form

$$\Pi_{t,\text{stock}}^{(k)} = q\left(S_t^{(k)} - S_t\right). \quad (22)$$

The total historical scenario PnL is obtained by summing over all active positions:

$$\Pi_t^{(k)} = \sum_{i=1}^{N_s} \Pi_{t,\text{stock},i}^{(k)} + \sum_{j=1}^{N_o} \Pi_{t,\text{option},j}^{(k)}. \quad (23)$$

This produces the empirical historical PnL distribution used for tail-risk estimation.

6.4 Weighted Historical VaR and ES

After constructing the historical scenario PnL series, the project computes VaR and ES from the empirical distribution. To allow more recent scenarios to receive greater influence, the historical engine uses exponentially decaying weights governed by a parameter $\lambda \in (0, 1)$. If there are W scenarios, the unnormalized weight of scenario k and the normalized weights are

$$\tilde{w}_k = (1 - \lambda)\lambda^{W-k}, \quad w_k = \frac{\tilde{w}_k}{\sum_{j=1}^W \tilde{w}_j}. \quad (24)$$

The scenarios are sorted from worst to best PnL, and the cumulative weighted distribution is used to determine the historical VaR. At confidence level α , VaR is the empirical weighted quantile corresponding to tail probability $1 - \alpha$. Expected Shortfall is then computed as the weighted average loss within the tail beyond the VaR threshold.

Formally, if $L_t^{(k)} = -\Pi_t^{(k)}$ denotes scenario loss, and if the losses are ordered from largest to smallest with corresponding normalized weights, then VaR is defined by the smallest threshold ℓ_α such that

$$\sum_{k: L_t^{(k)} \geq \ell_\alpha} w_k \leq 1 - \alpha, \quad (25)$$

and ES is the conditional weighted mean of the tail losses beyond this threshold. In implementation, the project computes these quantities numerically from the weighted empirical loss distribution.

6.5 One-Day-Scaled and True n -Day Historical Methods

The codebase supports two historical variants. The first is a true h -day historical simulation, in which shocks are computed over windows of length h and the full revaluation is carried out directly for that horizon. The second is a one-day historical method scaled to horizon h using the square-root-of-time rule. In that case, one first computes one-day historical VaR and ES, then scales them by $\sqrt{h}$.

The one-day-scaled method is computationally simpler and aligns with standard linear-risk approximations, but it is less theoretically appropriate for nonlinear option portfolios. The true h -day method preserves the horizon-specific historical shock construction and is therefore the more coherent nonparametric benchmark within the project.

6.6 Interpretation

The historical methodology implemented here should be understood as a hybrid between empirical scenario analysis and full instrument repricing. It remains nonparametric in the sense that shocks are drawn directly from realized market history, but it is not a simple return-based historical VaR applied to aggregate portfolio returns.

Instead, it reconstructs historical moves in the relevant market factors and then revalues the current portfolio under those shocks. This distinction is especially important for option portfolios, where nonlinear valuation and implied-volatility dynamics are central to the behavior of tail risk.

7 Monte Carlo Methodology

The Monte Carlo component of the risk engine estimates portfolio tail risk by simulating a large number of possible future market states and revaluing the current portfolio under each simulated outcome. Unlike the historical simulation framework, which reuses realized past shocks directly, the Monte Carlo approach imposes an explicit stochastic model for the evolution of the underlying assets and then approximates the future portfolio loss distribution by repeated random sampling.

Let t denote the forecast date, let h denote the risk horizon in trading days, and let M denote the number of Monte Carlo simulations. We generate a collection of simulated portfolio profit-and-loss values

$$\left\{ \Pi_t^{(1)}, \Pi_t^{(2)}, \dots, \Pi_t^{(M)} \right\}, \quad (26)$$

where each $\Pi_t^{(m)}$ corresponds to the horizon PnL of the current portfolio under one simulated path of future market factors. VaR and ES is then estimated from empirical distribution of simulated PnLs.

7.1 Underlying Price Dynamics

For equity underlyings, the project assumes that prices follow a geometric Brownian motion. If $S_{u,t}$ denotes the price of underlying u at time t , then its continuous-time dynamics are modeled as

$$\frac{dS_{u,t}}{S_{u,t}} = \mu_u dt + \sigma_u dW_{u,t}, \quad (27)$$

where μ_u is the drift, σ_u is the volatility, and $W_{u,t}$ is a standard Brownian motion.

For numerical simulation, the continuous-time process is discretized over daily time steps. If $\Delta t = 1/252$, then the one-step update for underlying u can be written as

$$S_{u,t+\Delta t} = S_{u,t} \exp \left[\left(\mu_u - \frac{1}{2} \sigma_u^2 \right) \Delta t + \sigma_u \sqrt{\Delta t} Z_u \right], \quad (28)$$

where $Z_u \sim \mathcal{N}(0, 1)$. Repeating this update over h trading days generates one simulated path from the forecast date to the horizon date.

7.2 Calibration of Drift, Volatility, and Correlation

The Monte Carlo engine requires estimates of the drift, volatility, and cross-asset dependence structure of the underlyings relevant to the portfolio. These parameters are calibrated from historical daily close data using the common aligned set of dates across the required equity universe.

Let $R_{u,\tau}$ denote the daily log return of underlying u on historical date τ ,

$$R_{u,\tau} = \ln \left(\frac{S_{u,\tau}}{S_{u,\tau-1}} \right). \quad (29)$$

From these historical returns, the code estimates the first and second moments needed for simulation. To place greater emphasis on more recent market observations, the calibration uses exponentially decaying weights controlled by a parameter $\lambda \in (0, 1)$. If the calibration window contains W observations, the unnormalized weight for observation k and the normalized weights are

$$\tilde{w}_k = (1 - \lambda)\lambda^{W-k}, \quad w_k = \frac{\tilde{w}_k}{\sum_{j=1}^W \tilde{w}_j}. \quad (30)$$

Using these weights, the weighted mean return and variance for underlying u is

$$\hat{\mu}_u^{(d)} = \sum_{k=1}^W w_k R_{u,k}, \quad \hat{\sigma}_u^{2,(d)} = \sum_{k=1}^W w_k \left(R_{u,k} - \hat{\mu}_u^{(d)} \right)^2. \quad (31)$$

The corresponding daily covariance between underlyings u and v is

$$\widehat{\text{Cov}}_{u,v}^{(d)} = \sum_{k=1}^W w_k \left(R_{u,k} - \hat{\mu}_u^{(d)} \right) \left(R_{v,k} - \hat{\mu}_v^{(d)} \right). \quad (32)$$

These daily estimates are then used to construct the covariance matrix and correlation matrix needed for multivariate simulation. The daily drift and volatility estimates can also be annualized when convenient using the 252-trading-day convention, although the simulation itself uses a daily time scale.

7.3 Cross-Asset Dependence and Correlated Shocks

Because the portfolio may contain multiple stocks and multiple option underlyings, it is necessary to simulate correlated market movements rather than independent price paths. Let $\Sigma^{(d)}$ denote the estimated daily covariance matrix of the aligned underlying return series. We use the Cholesky factorization with lower triangular matrix L to draw correlated shock vector ε with a random $Z \sim \mathcal{N}(0, I)$

$$\Sigma^{(d)} = LL^\top, \quad \varepsilon = LZ \quad (33)$$

The correlated shock vector has covariance matrix $\Sigma^{(d)}$. This transformation is applied at each simulated time step so that all underlyings evolve jointly under the estimated historical dependence structure.

The resulting Monte Carlo simulation is therefore multivariate: each path consists of a sequence of correlated daily shocks across all underlyings in the portfolio universe. This is important for realistic risk measurement because option portfolios often carry concentrated or offsetting exposure to multiple names, and the cross-sectional dependence among those names materially affects the loss distribution.

7.4 Simulated Horizon Valuation

Once the underlying paths have been generated, the portfolio is revalued at the horizon. For stock positions, the simulated value of a position with quantity q in underlying u under simulation m is simply

$$V_{u,t+h}^{(m)} = q S_{u,t+h}^{(m)}, \quad (34)$$

where $S_{u,t+h}^{(m)}$ is the simulated terminal stock price after h steps.

For option positions, the project reprices the contract using Black-Scholes at the simulated horizon state. If option j has strike K , option type τ , current expiry date T^{exp} , and underlying price $S_{t+h}^{(m)}$ under simulation m , then its horizon value is

$$O_{t+h}^{(m)} = BS\left(S_{t+h}^{(m)}, K, \tau_{t+h}, r_{t+h}, \sigma_{t+h}, \tau\right), \quad (35)$$

where τ_{t+h} is the remaining time to expiry at the horizon. In contrast to the current historical simulation implementation, the Monte Carlo approach naturally incorporates time decay by reducing time to maturity as the simulation moves forward. If the option expires before or at the horizon, then its value is taken to be intrinsic value rather than a Black-Scholes continuation value.

In the current implementation, the option volatility input used in Monte Carlo repricing is treated more simply than in the historical shock engine. The simulated underlyings are stochastic, while the implied-volatility treatment is anchored to the available surface representation used by the pricing layer. Thus the dominant source of randomness in Monte Carlo comes from simulated underlying-price dynamics and their dependence structure.

7.5 Portfolio Profit-and-Loss Distribution

Let V_t denote the current mark-to-market portfolio value at the forecast date and let $V_{t+h}^{(m)}$ denote the simulated portfolio value at horizon under path m . The scenario PnL and loss for simulation m is

$$\Pi_t^{(m)} = V_{t+h}^{(m)} - V_t, \quad L_t^{(m)} = -\Pi_t^{(m)}. \quad (36)$$

Collecting these across all M paths produces the empirical simulated loss distribution

$$\left\{L_t^{(1)}, L_t^{(2)}, \dots, L_t^{(M)}\right\}. \quad (37)$$

Monte Carlo VaR and ES are then estimated directly from this simulated distribution. If the losses are sorted in increasing order, then the VaR at level α is the empirical α -quantile of the simulated losses, and ES is the average of losses beyond that VaR threshold. If ℓ_α denotes the simulated VaR threshold,

$$\text{VaR}_\alpha = \ell_\alpha, \quad \text{ES}_\alpha = \mathbb{E}[L_t \mid L_t \geq \ell_\alpha], \quad (38)$$

with both quantities approximated empirically from the Monte Carlo sample.

7.6 Interpretation

The Monte Carlo methodology implemented here should be understood as a forward-looking scenario engine driven by a calibrated stochastic model of the underlying assets. Unlike historical simulation, which is tied to realized past shocks, Monte Carlo can generate a much larger scenario set and can produce future states not directly observed in the historical sample. At the same time, its accuracy depends on the

adequacy of the modeling assumptions: geometric Brownian motion for underlying prices, the weighted historical calibration of drift and covariance, and the treatment of option inputs within the repricing layer.

For nonlinear portfolios containing options, Monte Carlo provides an important complement to historical simulation. It preserves full portfolio revaluation while replacing empirical scenario replay with model-generated future paths, offering a distinct view of tail risk under the chosen stochastic assumptions.

8 Parametric Methodology

The parametric component of the risk engine estimates portfolio tail risk by imposing an explicit distributional approximation on the portfolio loss process. In contrast to historical simulation, which reuses realized shocks, and Monte Carlo simulation, which generates future market scenarios by repeated sampling from a stochastic process, the parametric methodology approximates the distribution of portfolio profit-and-loss through the first- and second-order sensitivities of the portfolio to the underlying risk factors. This makes the method computationally efficient, but its accuracy depends on the adequacy of the local approximation and the distributional assumptions used for the underlying factors.

Let t denote the forecast date and let h denote the risk horizon in trading days. The parametric framework begins by specifying a vector of market risk factors X_t that drive the portfolio value. For our use case, the dominant factors are the underlying equity prices associated with the stock positions and the underlyings of option positions. The portfolio value is therefore viewed as a function $V_t = V(X_t)$ and the aim is to approximate the distribution of the horizon profit-and-loss

$$\Pi_{t,t+h} = V(X_{t+h}) - V(X_t). \quad (39)$$

Rather than revaluing the full portfolio under a large set of scenarios, the parametric method approximates this change analytically using local sensitivities and an assumed distribution for factor changes.

8.1 Delta Approximation of Portfolio PnL

The first-order approximation treats the portfolio value as locally linear in the underlying risk factors. Let $X_t = (X_{1,t}, \dots, X_{n,t})^\top$ denote the vector of relevant factor levels and let $\Delta X = X_{t+h} - X_t$ denote the horizon change. With first-order Taylor approximation, the horizon PnL is approximation by

$$V(X_{t+h}) \approx V(X_t) + \nabla V(X_t)^\top \Delta X, \quad \Pi_{t,t+h} \approx \nabla V(X_t)^\top \Delta X. \quad (40)$$

The vector $\nabla V(X_t)$ is the portfolio sensitivity vector with respect to the chosen factors. In the context of equity and option portfolios, these sensitivities consist primarily of stock exposures and option deltas with respect to their underlyings. Thus the parametric methodology reduces portfolio risk estimation to the distributional analysis of a linear combination of factor shocks.

For a stock position with quantity q and underlying price S_t , sensitivity to the underlying is simply

$$\frac{\partial V^{(s)}}{\partial S_t} = q. \quad (41)$$

For an option position with signed quantity q , underlying price S_t , strike K , time to maturity τ_t , risk-free rate r_t , and implied volatility σ_t , the first-order sensitivity to the underlying is given by the Black-Scholes delta, with different formulas for calls and puts

$$\frac{\partial V^{(o)}}{\partial S_t} = q \Delta_t, \quad \Delta_t^{\text{call}} = \Phi(d_1), \quad \Delta_t^{\text{put}} = \Phi(d_1) - 1. \quad (42)$$

Here d_1 is the standard Black-Scholes quantity

$$d_1 = \frac{\ln(S_t/K) + (r_t + \frac{1}{2}\sigma_t^2)\tau_t}{\sigma_t\sqrt{\tau_t}}. \quad (43)$$

These sensitivities are evaluated at the current market state and then aggregated across all positions to form the total exposure of the portfolio to each underlying factor.

8.2 Quadratic Correction and Gamma Effects

Because option values are nonlinear in the price of their underlying asset, a purely first-order approximation may understate or mischaracterize the effect of larger market moves. The parametric framework therefore augments the linear exposure of the portfolio with a second-order correction based on option gamma. Rather than constructing the full Hessian of the portfolio with respect to all market factors, we use underlying-level Black-Scholes gamma exposures and aggregates them across positions.

For an option position, the first-order sensitivity to the underlying price is its Black-Scholes delta, while the second-order sensitivity is its Black-Scholes gamma. If an option has current underlying price S_t , strike K , implied volatility σ_t , and time to maturity τ_t , then its gamma is

$$\Gamma_t = \frac{\partial^2 V^{(o)}}{\partial S_t^2} = \frac{\phi(d_1)}{S_t \sigma_t \sqrt{\tau_t}}, \quad (44)$$

where $\phi(\cdot)$ denotes the standard normal density and d_1 is the standard Black-Scholes quantity. This gamma term captures the local curvature of option value with respect to its underlying and therefore provides a second-order correction to the portfolio's risk exposure.

In implementation, the parametric engine does not evaluate a full quadratic form based on the complete Hessian matrix. Instead, it computes the delta and gamma of each option position at the current market state, converts them into dollar exposures at the level of each underlying ticker, and then aggregates these exposures across all stock and option positions in the portfolio. Stocks contribute only to the first-order exposure, while options contribute to both first- and second-order exposure.

This construction leads to a delta-gamma approximation in which the portfolio is summarized by an underlying-level first-order exposure vector and an underlying-level second-order exposure vector. These objects are then combined with the calibrated drift, volatility, and covariance structure of the underlying assets to produce an approximate portfolio mean and variance over the risk horizon. Compared with full repricing, this methodology is substantially faster, but it remains an approximation because higher-order nonlinearities, explicit implied-volatility shocks, and path-dependent effects are not modeled directly.

8.3 Factor Shock Distribution

The parametric method requires a distributional assumption for the factor shocks ΔX . The project assumes that factor returns are jointly Gaussian over the horizon, with mean vector and covariance matrix estimated from historical data. Let R_t denote the vector of daily underlying returns, and let $\mu^{(d)}$ and $\Sigma^{(d)}$ denote the corresponding daily mean vector and covariance matrix estimated from the historical window. The estimation uses exponentially decaying weights controlled by a parameter $\lambda \in (0, 1)$.

For W historical observations, the unnormalized and normalized weight assigned to observation k is

$$\tilde{w}_k = (1 - \lambda)\lambda^{W-k}, \quad w_k = \frac{\tilde{w}_k}{\sum_{j=1}^W \tilde{w}_j}. \quad (45)$$

The weighted mean return vector and covariance matrix is therefore

$$\hat{\mu}^{(d)} = \sum_{k=1}^W w_k R_k, \quad \hat{\Sigma}^{(d)} = \sum_{k=1}^W w_k (R_k - \hat{\mu}^{(d)})(R_k - \hat{\mu}^{(d)})^\top. \quad (46)$$

To move from daily to horizon- h quantities, the standard square-root-of-time scaling is used. The horizon mean vector and covariance matrix is approximated as

$$\hat{\mu}^{(h)} = h\hat{\mu}^{(d)}, \quad \hat{\Sigma}^{(h)} = h\hat{\Sigma}^{(d)}. \quad (47)$$

Under the parametric assumption, the factor-return vector over the horizon is therefore modeled as

$$R_{t,t+h} \sim \mathcal{N}(\hat{\mu}^{(h)}, \hat{\Sigma}^{(h)}). \quad (48)$$

This assumption is the main probabilistic input into the parametric VaR and ES calculations.

8.4 Portfolio Delta and Gamma Exposures

Let the relevant underlying tickers in the portfolio be indexed by $i = 1, \dots, n$. For each ticker i , the code constructs a first-order dollar exposure D_i and a second-order dollar exposure G_i . These quantities are defined at the level of the underlying stock rather than at the level of the individual instrument.

For stock positions, the contribution is purely first-order. If the portfolio holds a signed quantity $q_i^{(s)}$ of stock i at current price S_i , then the stock contribution to first-order exposure is

$$D_i^{\text{stock}} = q_i^{(s)} S_i. \quad (49)$$

Stocks do not contribute to second-order exposure, so their gamma contribution is zero.

For options written on underlying i , the code first computes the Black-Scholes delta and gamma of each contract at the current market state. If option j has quantity $q_j^{(o)}$, delta Δ_j , gamma Γ_j , and underlying price S_i , then its contribution to the first-order and second-order dollar exposures is

$$D_i^{\text{option},j} = q_j^{(o)} \Delta_j S_i, \quad G_i^{\text{option},j} = q_j^{(o)} \Gamma_j S_i^2. \quad (50)$$

Summing across all stock and option positions tied to underlying i gives the total exposure vectors

$$D_i = D_i^{\text{stock}} + \sum_{j: \text{underlying}(j)=i} D_i^{\text{option},j}, \quad G_i = \sum_{j: \text{underlying}(j)=i} G_i^{\text{option},j}. \quad (51)$$

In vector form, we represent the current portfolio by two n -dimensional exposure vectors,

$$D = (D_1, \dots, D_n)^\top, \quad G = (G_1, \dots, G_n)^\top. \quad (52)$$

D captures the portfolio's first-order dollar sensitivity to underlying price changes, while G captures the local curvature of the option component of the portfolio with respect to the same underlying prices.

8.5 Horizon Scaling of Calibrated Inputs

Let $\mu = (\mu_1, \dots, \mu_n)^\top$ denote the calibrated drift vector, $\sigma = (\sigma_1, \dots, \sigma_n)^\top$ the calibrated volatility vector, and ρ the calibrated correlation matrix of the underlying assets. The corresponding covariance matrix is constructed for every component as

$$\Sigma = \text{diag}(\sigma) \rho \text{diag}(\sigma), \quad \Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}. \quad (53)$$

For h -day risk horizon, the project uses the 252-trading-day convention and sets $\Delta t = \frac{h}{252}$.

The horizon-adjusted drift term and covariance matrix used in the parametric approximation is

$$\mu^{(h)} = \left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t, \quad \Sigma^{(h)} = \Sigma \Delta t. \quad (54)$$

$\mu^{(h)}$ is the geometric-Brownian-motion drift correction appearing in the log-price dynamics, while $\Sigma^{(h)}$ is the covariance structure of the horizon underlying shocks used in the portfolio approximation.

8.6 Portfolio Mean Approximation

The parametric engine approximates portfolio profit-and-loss by expanding each position value locally in the underlying price change. For an option position, a second-order Taylor approximation gives

$$\Delta V \approx \Delta \Delta S + \frac{1}{2} \Gamma (\Delta S)^2, \quad (55)$$

where Δ is the Black-Scholes delta, Γ is the Black-Scholes gamma, and ΔS is the change in the underlying price over the horizon. Multiplying by option quantity q , the position-level approximation is

$$\Delta V^{(o)} \approx q \Delta \Delta S + \frac{1}{2} q \Gamma (\Delta S)^2. \quad (56)$$

For stock positions, only the first-order term is present, so if a stock position has quantity q then

$$\Delta V^{(s)} \approx q \Delta S. \quad (57)$$

To connect this approximation with calibrated return model, the code writes price changes in relative form. If S_i is the current price of underlying i and R_i denotes its horizon return shock, we get

$$\Delta S_i \approx S_i R_i, \quad \Delta V^{(o)} \approx q \Delta S_i R_i + \frac{1}{2} q \Gamma S_i^2 R_i^2. \quad (58)$$

This motivates the two exposure quantities used in the code, as discussed in section 8.4. For each underlying ticker i , the first-order dollar exposure is defined by aggregating stock and option delta contributions and the second-order dollar exposure is defined by aggregating option gamma contributions,

$$D_i = q_i^{(s)} S_i + \sum_{j: \text{underlying}(j)=i} q_j^{(o)} \Delta_j S_i, \quad G_i = \sum_{j: \text{underlying}(j)=i} q_j^{(o)} \Gamma_j S_i^2 \quad (59)$$

In vector form, the portfolio is therefore represented by the underlying-level exposure vectors

$$D = (D_1, \dots, D_n)^\top, \quad G = (G_1, \dots, G_n)^\top. \quad (60)$$

Taking expectations under the calibrated horizon return model yields the approximate portfolio mean. The first-order contribution is

$$\mu_\Delta = D^\top \mu^{(h)}, \quad (61)$$

where $\mu^{(h)}$ is the horizon drift vector. The second-order contribution comes from the fact that

$$\mathbb{E}[(\Delta S_i)^2] \approx S_i^2 \mathbb{E}[R_i^2]. \quad (62)$$

Under GBM-based horizon approximation, this second moment is represented by $\sigma_i^2 \Delta t$, where σ_i is the calibrated underlying volatility and $\Delta t = h/252$. The gamma contribution to the portfolio mean is

$$\mu_\Gamma = \frac{1}{2} \sum_{i=1}^n G_i \sigma_i^2 \Delta t. \quad (63)$$

Combining the two pieces gives the portfolio mean approximation

$$\mu_P = \mu_\Delta + \mu_\Gamma = D^\top \mu^{(h)} + \frac{1}{2} \sum_{i=1}^n G_i \sigma_i^2 \Delta t. \quad (64)$$

Thus the portfolio mean is not driven solely by directional exposure. The second-order correction accounts for the fact that option curvature causes expected PnL to depend on the variance of the underlying move, even when the expected return itself is small.

8.7 Portfolio Variance Approximation

The variance approximation follows the same delta-gamma logic. The linear component of portfolio PnL is the weighted combination of underlying return shocks described by the exposure vector D , so its variance is the standard quadratic form

$$\sigma_\Delta^2 = D^\top \Sigma^{(h)} D, \quad (65)$$

where $\Sigma^{(h)}$ is the horizon covariance matrix of the underlying return shocks.

To incorporate nonlinear curvature effects, the code adds a second-order gamma variance term. Since the quadratic contribution for each underlying is proportional to $G_i R_i^2$, the variance of the second-order part depends on products of squared covariance terms. In the implementation, this is approximated by

$$\sigma_\Gamma^2 = \frac{1}{2} G^\top \left(\Sigma^{(h)} \circ \Sigma^{(h)} \right) G, \quad (66)$$

where $\circ$ denotes elementwise multiplication. This term is the code-level approximation used to capture the contribution of option curvature to total portfolio variance.

The resulting portfolio variance and standard deviation is therefore

$$\sigma_P^2 = \sigma_\Delta^2 + \sigma_\Gamma^2 = D^\top \Sigma^{(h)} D + \frac{1}{2} G^\top \left(\Sigma^{(h)} \circ \Sigma^{(h)} \right) G, \quad \sigma_P = \sqrt{\sigma_P^2}. \quad (67)$$

This construction gives the parametric engine a second-order correction for option nonlinearity while preserving computational tractability. The variance is therefore driven by both linear underlying exposure and the additional dispersion introduced by gamma-sensitive option positions.

8.8 Normal Approximation and Closed-Form Tail Risk

Once the portfolio mean μ_P and standard deviation σ_P have been computed, the project assumes that the approximate portfolio PnL is Gaussian and the equivalent portfolio loss is modeled as normally distributed with mean $-\mu_P$ and variance σ_P^2 .

$$\Pi_{t,t+h} \sim \mathcal{N}(\mu_P, \sigma_P^2), \quad L_{t,t+h} = -\Pi_{t,t+h} \quad (68)$$

For $z_\alpha = \Phi^{-1}(1 - \alpha)$, where $\Phi^{-1}(\cdot)$ is the inverse standard normal cumulative distribution function. Since α is typically large (for example, 0.95 or 0.99), the quantity z_α is negative in the implementation.

The reported parametric VaR and Expected Shortfall is then

$$\text{VaR}_\alpha = -(\mu_P + z_\alpha \sigma_P), \quad \text{ES}_\alpha = -\left(\mu_P - \frac{\phi(z_\alpha)}{1 - \alpha} \sigma_P\right), \quad (69)$$

where $\phi(\cdot)$ denotes the standard normal density.

8.9 Interpretation

The parametric methodology implemented here should be understood as a sensitivity-based approximation to the portfolio loss distribution. Relative to Monte Carlo or historical simulation, it is computationally inexpensive because it compresses the portfolio into a small set of underlying-level delta and gamma exposures and then combines those exposures with analytically tractable moment estimates. This makes it especially useful as a fast benchmark or approximation method.

At the same time, the method is also the most model-dependent of the three approaches in the project. It assumes that underlying factor shocks are adequately summarized by their calibrated mean, volatility, and correlation structure, and that the portfolio loss can be represented accurately through a delta-gamma expansion combined with a normal approximation. For portfolios with substantial nonlinear exposure, these assumptions should be interpreted as approximate rather than exact. Nevertheless, the methodology provides a coherent and computationally efficient parametric benchmark against which the historical and Monte Carlo engines can be compared.

9 Backtesting Framework

The project evaluates the empirical quality of its risk forecasts by comparing predicted portfolio tail-risk measures against the losses that are subsequently realized over the chosen holding period. On each forecast date t , the risk engine computes portfolio VaR and ES using only the information available at that time. The portfolio is then held fixed at date t and revalued over the forward horizon $[t, t + h]$ in order to obtain realized profit-and-loss. Repeating this procedure across a sequence of forecast dates produces a time series of model forecasts and realized outcomes for each risk methodology.

The realized horizon profit-and-loss is constructed at the portfolio level rather than from a purely return-based approximation. Stock positions are revalued using the observed forward price change, while option positions are revalued using the project's option-pricing framework, with consistent treatment of contracts that expire within the backtest horizon. This ensures that the backtest compares each forecast against a realized outcome defined under the same valuation architecture used throughout the risk engine.

The primary object of interest in the backtest is the VaR exceedance event. If the realized portfolio loss over the horizon exceeds the predicted VaR threshold, then the forecast is said to be breached. Over time, the observed frequency of such breaches can be compared with the theoretical exceedance probability $1 - \alpha$. This provides a direct measure of model calibration: a method that produces too many exceedances may be understating tail risk, while a method that produces too few may be overly conservative. Expected Shortfall is also recorded and compared to realized losses, but it is interpreted as a complementary tail-severity benchmark rather than as the main object of formal exceedance testing.

9.1 Statistical Evaluation

To assess whether the realized exceedance frequency is consistent with the forecast confidence level, the project applies the Kupiec Proportion of Failures (POF) test. Let N denote the number of backtest observations and let x denote the number of VaR exceedances observed over those forecasts. Under the null hypothesis, the exceedance probability is exactly

$$p = 1 - \alpha. \quad (70)$$

If the risk model is correctly calibrated, then the observed exceedance count should be statistically consistent with a Bernoulli sequence with failure probability p .

Let the realized exceedance rate be

$$\hat{p} = \frac{x}{N}. \quad (71)$$

The likelihood of the observed exceedance sequence under the null hypothesis is

$$L_0 = (1 - p)^{N-x} p^x, \quad (72)$$

while the likelihood under the unrestricted alternative is

$$L_1 = (1 - \hat{p})^{N-x} \hat{p}^x. \quad (73)$$

The Kupiec likelihood-ratio statistic is then

$$LR_{\text{POF}} = -2 \ln \left(\frac{L_0}{L_1} \right), \quad LR_{\text{POF}} = -2 \left[(N - x) \ln \left(\frac{1 - p}{1 - \hat{p}} \right) + x \ln \left(\frac{p}{\hat{p}} \right) \right]. \quad (74)$$

Under the null hypothesis, this statistic is asymptotically distributed as a chi-squared random variable with one degree of freedom. The corresponding p -value is therefore obtained from the $\chi^2(1)$ distribution and used to determine whether the realized failure rate is statistically plausible under the model.

Because the Kupiec test can be strict in finite samples, especially when backtest windows overlap or when the number of observations is limited, the project also reports an exact binomial coverage test. In that case, the observed exceedance count x is compared directly against the binomial distribution

$$X \sim \text{Binomial}(N, 1 - \alpha), \quad (75)$$

which provides an exact finite-sample calibration benchmark under the same null exceedance probability. The summary output reports the corresponding exact binomial p -value together with a confidence interval for the realized exceedance rate.

In addition, the framework includes an interval-based traffic-light classification for VaR exceedances. This classification maps the realized exceedance count into a qualitative region, such as benign, warning, or adverse, based on its location relative to reference exceedance thresholds under the same binomial null model. Taken together, these diagnostics allow the backtesting framework to distinguish between severe model misspecification and milder forms of over- or under-conservatism.

9.2 Backtesting Outputs

The backtesting framework produces both tabular and graphical outputs. At the tabular level, it records the number of observations, the expected and realized number of exceedances, the corresponding exceedance percentages, and the results of the statistical calibration tests for each methodology. The summary tables therefore report not only whether a model is accepted or rejected under a given test, but also how far the realized exceedance behavior deviates from its nominal target.

At the graphical level, the framework generates several complementary diagnostics. Rolling exceedance-count plots summarize the local clustering of breaches over time, hit-sequence plots show the exact timing of exceedance events, realized-loss-versus-threshold plots compare realized losses directly with the forecast VaR or ES series, and exceedance scatter plots summarize the magnitude of breaches relative to the model-implied threshold. Together, these outputs describe not only how often each model fails, but also when those failures occur and how large they are when they do occur.

9.3 Why Expected Shortfall Is Not Tested in the Same Way

Expected Shortfall is recorded throughout the backtesting framework, but it is not treated as the primary object of formal hit-based testing in the same way as VaR. The reason is that VaR defines a threshold event: on each forecast date, the realized loss either exceeds the VaR level or it does not. This naturally produces a binary exceedance sequence, which can be analyzed statistically using failure-rate tests such as the Kupiec POF procedure.

Expected Shortfall, by contrast, is a conditional tail expectation rather than a quantile threshold. It measures the average loss given that the loss has already exceeded the VaR boundary, and therefore does not by itself generate a simple binary event sequence. As a result, there is no directly analogous exceedance count for ES that can be evaluated using the same Bernoulli or binomial framework.

In the present project, ES is therefore treated primarily as a complementary tail-severity measure. The code records whether realized losses exceed the predicted ES level and includes ES in the tabular and graphical diagnostics, but these comparisons should be interpreted as descriptive rather than as formal calibration tests of the same type used for VaR. In other words, ES is useful for understanding the magnitude of tail losses, while VaR remains the main object of formal exceedance-based backtesting in this framework.

10 Results

10.1 Overview and Experimental Design

The empirical analysis compares the historical, Monte Carlo, and parametric risk engines across portfolios with different exposure profiles and under different market conditions. The objective is not only to compare the size of the reported VaR and ES estimates, but also to evaluate how well each methodology captures the realized frequency and severity of subsequent portfolio losses.

For each scenario, the project generates a time series of VaR and ES forecasts, extends those forecasts with realized forward profit-and-loss, and evaluates the resulting exceedance behavior using statistical tests and graphical diagnostics. The main emphasis is on VaR calibration, since VaR defines a natural exceedance event that can be tested formally, while ES is interpreted primarily as a complementary tail-severity measure.

The experiments are organized along two dimensions: portfolio structure and market environment. Portfolios are grouped into three categories: equity-only, options-only, and mixed equity–option portfolios. Market environments range from calmer periods to recent high-volatility episodes and more severe stress regimes such as the 2008 financial crisis. This structure makes it possible to study how model performance changes with both portfolio nonlinearity and market stress.

10.2 Scenario Summary

The equity-only scenarios establish the baseline behavior of the models under diversified, concentrated, and single-name stock exposure across calm, volatile, and crisis periods. The options-only scenarios isolate nonlinear exposure by comparing long-gamma and short-volatility portfolios under a common local stress regime. The mixed equity–option scenario provides an intermediate case in which linear equity exposure and nonlinear option exposure coexist in the same portfolio.

10.2.1 Equity-Only Scenarios

Scenario	Regime	Purpose
long_equity_core_calm_regime	Calm	Diversified equity baseline
long_equity_core_stress_regime	2020–2021 stress	Equity calibration under stress
long_equity_core_gfc_regime	2008–2009 crisis	Equity behavior in crisis
tech_concentrated_equity_2021_vol_regime	2021–2022 volatile	Concentrated equity risk
single_name_aapl_gfc_regime	2008–2009 crisis	AAPL single-name crisis risk
single_name_msft_gfc_regime	2008–2009 crisis	MSFT single-name crisis risk
single_name_aapl_2021_vol_regime	2021–2022 volatile	AAPL single-name volatile regime
single_name_msft_2021_vol_regime	2021–2022 volatile	MSFT single-name volatile regime

Table 1: Equity-only backtesting scenarios.

10.2.2 Options-Only Scenarios

The options-only experiments are designed to isolate nonlinear exposure under a common stress regime. The set includes long-convexity portfolios, short-volatility portfolios, downside-hedge portfolios, and directional spread portfolios, so that model behavior can be compared across several distinct option payoff shapes.

Scenario	Regime	Purpose
long_gamma_tech_stress_regime	Local stress	Long-convexity multi-name option risk
short_vol_pair_stress_regime	Local stress	Short-vol tail-risk behavior
long_straddle_aapl_stress_regime	Local stress	Pure long-vol convexity on a single name
short_strangle_tech_stress_regime	Local stress	Dispersed short-volatility exposure
protective_put_basket_stress_regime	Local stress	Downside-hedge style put exposure
bear_put_spread_aapl_stress_regime	Local stress	Directional bearish spread exposure

Table 2: Options-only backtesting scenarios.

10.2.3 Equity–Option Combination Scenarios

The mixed portfolio experiments are intended to study portfolios in which linear equity exposure and nonlinear option exposure coexist in the same position set. Because the local option-surface dataset is currently available only for the 2026 window, these scenarios are organized across two local subperiods: an early stress regime and a later transition regime. Within that constraint, the experiment set spans several distinct hedged structures, including covered-call overlays, protective hedges, collars, and approximately delta- and gamma-neutral constructions.

Scenario	Regime	Purpose
covered_call_aapl_local_regime	Early local stress	Covered-call overwrite exposure
protective_put_aapl_local_stress_regime	Early local stress	Downside hedge with retained equity upside
collar_aapl_local_stress_regime	Early local stress	Capped-upside, protected-downside hedge
delta_neutral_conversion_aapl_local_stress_regime	Early local stress	Approx. delta/gamma-neutral stock-option hedge
reverse_conversion_msft_local_transition_regime	Local transition	Approx. delta/gamma-neutral short-stock hedge
tech_collar_basket_local_transition_regime	Local transition	Multi-name hedged equity basket

Table 3: Equity–option combination backtesting scenarios.

10.3 Equity-Only Portfolio Results

The equity-only backtests produce a clear ranking among the four VaR methodologies. Across most scenarios, the historical n -day model is the best calibrated, with realized exceedance rates closest to the nominal 5% target and the strongest overall statistical acceptance. Monte Carlo is generally the next-best performer, especially in the more recent stress and high-volatility regimes. The parametric model serves as a useful benchmark but is often somewhat conservative. By contrast, the 1-day-scaled historical method consistently produces exceedance rates that are too low, indicating that square-root-of-time scaling tends to overstate risk even for stock-only portfolios.

Performance also depends materially on the market regime. In the calm diversified-equity setting, the historical n -day model performs best, while the remaining methods are conservative. In the 2020–2021 stress period, historical n -day, Monte Carlo, and parametric VaR all perform reasonably well. In the 2008–2009 crisis regime, however, all four methods become conservative, with exceedance rates well below target. The same pattern broadly appears in the single-name and concentrated-equity experiments.

These outcomes can be summarized briefly as follows:

- **long_equity_core_calm_regime:** Historical n -day performs well; other methods are conservative.
- **long_equity_core_stress_regime:** Historical n -day, Monte Carlo, and parametric all perform reasonably well; 1-day scaling remains weak.
- **long_equity_core_gfc_regime:** All methods are overly conservative in the crisis window.

- **tech_concentrated_equity_2021_vol_regime:** Historical n -day and Monte Carlo perform best; parametric is acceptable but conservative.
- **single_name_aapl_2021_vol_regime:** Historical n -day performs well; others are conservative.
- **single_name_aapl_gfc_regime:** Historical n -day, Monte Carlo, parametric perform reasonably.
- **single_name_msft_2021_vol_regime:** Historical n -day and Monte Carlo perform well; parametric is somewhat conservative.
- **single_name_msft_gfc_regime:** All methods are strongly conservative; weak calibration overall.

Expected Shortfall results are materially weaker than the VaR results across the equity-only experiments. ES exceedance rates are generally well below the nominal level, and formal statistical acceptance is rare. Overall, the equity-only results suggest that the models perform reasonably well for VaR forecasting when using the historical n -day methodology, with Monte Carlo as a credible second-best approach. The parametric model is usable but usually conservative, while the 1-day-scaled historical method does not perform well and is consistently too conservative.

10.4 Options-Only Portfolio Results

The options-only backtests are materially less stable than the equity-only results, which is consistent with the greater nonlinearity of pure option portfolios. Unlike the equity-only cases, there is no single method that dominates across all scenarios. Performance depends strongly on the payoff shape of the portfolio and on whether the strategy is long convexity, short volatility, or directionally structured. In some cases, the historical methods remain closest to the target exceedance rate, while in others Monte Carlo and parametric VaR materially understate risk. Overall, the options-only results show that model calibration becomes much more sensitive once the portfolio is driven primarily by convexity, volatility exposure, and asymmetric payoff structure.

A useful distinction emerges between portfolios with relatively bounded or structured downside and portfolios with strong convexity or short-vol tail exposure. The short strangle and bear put spread perform comparatively well, with several methods producing acceptable VaR calibration. By contrast, the long-gamma technology portfolio and the short-volatility pair both produce systematically poor VaR calibration, with exceedance rates well above the nominal 5% level. The long straddle and protective put basket lie between these extremes: some methods remain acceptable, but the stronger revaluation-based models often become too aggressive and generate too many breaches. These outcomes are:

- **long_gamma_tech_stress_regime:** This portfolio is long call and put exposure across different underlying names, so it is strongly convex and highly sensitive to both large spot moves and volatility repricing. All VaR methods perform poorly, suggesting that the realized loss behavior of this multi-name long-gamma structure is not captured well by any of the models.
- **short_vol_pair_stress_regime:** This portfolio is short option premium through short call and short put positions, so it has negative convexity and substantial tail exposure. All VaR methods are rejected, indicating that the models underestimate the loss risk associated with short-volatility payoffs during stress.
- **long_straddle_aapl_stress_regime:** The strategy holds both a long at-the-money call and a long at-the-money put on AAPL, making it a pure long-volatility and long-convexity trade. Historical 1-day-scaled and parametric VaR perform reasonably well, while historical n -day and Monte Carlo

are too aggressive, suggesting that the more complex scenario-based methods may overreact to realized move and volatility patterns in this case.

- **protective_put_basket_stress_regime:** This portfolio consists of long downside put exposure on multiple names, so losses are naturally more structured and partially limited relative to open-ended short-volatility trades. Historical 1-day-scaled VaR performs best, while the other methods are too aggressive, indicating that the downside-hedge style payoff produces more stable tail behavior than broader convex long-gamma strategies.
- **bear_put_spread_aapl_stress_regime:** This strategy combines a long put with a short lower-strike put, creating a bearish directional spread with capped upside and capped downside. The historical methods perform well because the spread structure limits extreme tail exposure, whereas Monte Carlo and parametric VaR understate risk, likely because they do not fully capture how realized losses cluster within the spread’s nonlinear payoff region.
- **short_strangle_tech_stress_regime:** This portfolio sells out-of-the-money calls and puts, collecting premium while accepting bounded but dispersed downside from large underlying moves. It is the strongest options-only case, with all four VaR methods performing reasonably well, likely because the strikes are farther from spot and the payoff is less immediately exposed than the more concentrated short-volatility pair.

Expected Shortfall behaves somewhat better in the options-only experiments than it did in the equity-only group, but the interpretation remains uneven across portfolio types. ES calibration is acceptable in some bounded or structured portfolios, such as the short strangle and bear put spread, but deteriorates in the more strongly convex long-gamma cases.

Overall, the options-only results suggest that none of the methods is uniformly reliable for pure option portfolios. Historical methods are often the most robust, but the broader conclusion is that model performance is highly scenario-dependent and generally weaker than in the equity-only setting.

10.5 Equity–Option Combination Portfolio Results

The equity–option combination portfolios occupy an intermediate position between the equity-only and options-only cases. In principle, these portfolios should benefit from partial hedging, capped payoff regions, or approximate neutrality, and therefore produce more stable risk behavior than pure option portfolios. In practice, however, the results depend strongly on the exact hedge design. Some mixed portfolios behave conservatively and produce very few breaches, while others still exhibit substantial exceedance rates despite being constructed to reduce directional or convexity exposure.

A useful distinction emerges between the early local stress AAPL overlays and the later transition-regime hedged structures. The covered call, protective put, collar, and conversion-style AAPL portfolios all produce very similar backtesting outcomes. In each case, the historical methods record four VaR breaches and the Monte Carlo and parametric methods record three, implying realized exceedance rates of approximately 11.8% and 8.8%, respectively. This suggests that, over this short stressed sample, the common AAPL price path dominates the differences among the hedge structures. In other words, although the payoff profiles of these portfolios differ, the realized tail events occur on broadly the same dates, so the backtest distinguishes them less sharply than might be expected from their static definitions.

The later transition-regime portfolios show much greater dispersion. The reverse-conversion-style MSFT portfolio performs poorly across all VaR methodologies, with realized exceedance rates ranging from 14.6% to 29.3%. This indicates that the approximate neutrality of the structure does not translate into stable realized tail behavior. Residual directional exposure, imperfect hedge balance, and nonlinear repricing effects appear to remain large enough for all four models to understate risk materially.

By contrast, the hedged technology collar basket is the strongest mixed-portfolio case. Its realized VaR exceedance rates range from 0% to 2.4%, implying that the combination of diversification and collar protection substantially suppresses realized tail losses. The models appear conservative rather than aggressive. These outcomes can be summarized briefly as follows:

- **covered_call_aapl_local_regime:** The portfolio remains materially exposed to downside equity risk, so all methods produce exceedance rates above target despite the short-call overlay.
- **protective_put_aapl_local_stress_regime:** The downside hedge improves payoff structure, but breach frequency remains similar to the covered-call case over the short stress sample.
- **collar_aapl_local_stress_regime:** The collar creates a more bounded payoff, yet the realized exceedance pattern remains close to the other AAPL overlay portfolios.
- **delta_neutral_conversion_aapl_local_stress_regime:** Approximate neutrality does not eliminate realized breaches, suggesting that residual repricing and path effects remain important.
- **reverse_conversion_msft_local_transition_regime:** This is the weakest mixed-portfolio case; all methods materially understate risk despite the intended hedge structure.
- **tech_collar_basket_local_transition_regime:** This is the strongest mixed-portfolio case; diversification and collar protection produce a very conservative realized loss profile.

Expected Shortfall behaves somewhat more favorably in the mixed portfolios than in the equity-only experiments, particularly for the more structured hedge portfolios. In the collar-type cases, ES exceedance rates are generally acceptable, while the reverse-conversion portfolio remains weak under both VaR and ES. Overall, the combined-portfolio results suggest that hedge structure matters more than the simple presence of options. Portfolios with diversified and explicitly bounded downside behavior can be modeled reasonably conservatively, while portfolios that are only approximately neutral may still backtest poorly if residual pathwise and nonlinear exposure remains large.

10.6 Detailed Backtesting Results and Plots

Exact results can be found under the `results` folder of the uploaded GitHub repository, where each scenario includes the full forecast output, realized forward PnL, exceedance backtests, summary tables of VaR and ES calibration metrics, and diagnostic plots such as rolling exceedance counts, hit sequences, realized-loss-versus-threshold comparisons, and exceedance scatter plots.

Link to results folder in the github repo:

<https://github.com/AhaanKanaujia/risk-engine/tree/main/results>

11 Computational Architecture

The project is organized as a modular research codebase separating data access, valuation, risk-model construction, backtesting, and analysis. This keeps the market-data layer, pricing layer, scenario-generation layer, and evaluation layer logically distinct, so that changes to one component do not require rewriting the entire system. In practice, the codebase is divided into modules for portfolio representation, Black-Scholes pricing, implied-volatility handling, forecasting, backtesting, and visualization.

A shared cache layer is used for historical prices, risk-free rates, daily option surfaces, processed implied-volatility data, and interpolator objects, which reduces repeated file reads and repeated surface reconstruction. Performance is further improved through batch implied-volatility inversion, reusable aligned market-data histories, and a hybrid numerical procedure that combines Newton iteration with Brent fallback when needed.

At the workflow level, the engine is staged clearly into forecast generation, realized-PnL construction, exceedance testing, and output reporting. This makes the system easier to debug, validate, and extend, while preserving a clear connection between model inputs, portfolio valuation logic, and final risk diagnostics.

12 Assumptions and Limitations

The engine values options using the Black-Scholes framework and relies on standard diffusion-based assumptions for equity dynamics. The parametric model assumes a Gaussian structure for factor returns, while the Monte Carlo engine uses geometric Brownian motion with historically calibrated parameters. These choices make the system tractable, but they do not fully capture jumps, fat tails, volatility clustering, transaction costs, or other market frictions that may become important during stress.

Implied-volatility surfaces are inferred from daily option close prices and then interpolated or extrapolated as needed for valuation and historical shock construction. Historical simulation uses fixed-expiry volatility shocks to reduce maturity roll-down effects, but this is still an approximation to the full dynamics of the volatility surface. Monte Carlo and parametric methods also treat volatility risk more simply than a fully structural stochastic-volatility framework would.

The entire framework operates on daily data under a 252-trading-day convention, so intraday effects, liquidity conditions, and higher-frequency behavior are not modeled directly. Each methodology also introduces approximation error: historical simulation applies past shocks to the current portfolio rather than replaying realized historical portfolio values, the parametric model uses a local delta-gamma approximation, and the 1-day-scaled historical method relies on square-root-of-time scaling. Finally, the backtesting framework is most naturally suited to VaR exceedance analysis; ES is more difficult to test formally in the same way, and overlapping forecast windows reduce independence across observations.

13 Project Outcome

The project succeeds in building a coherent portfolio risk engine for equities and European options, with full mark-to-market valuation, multiple VaR and ES methodologies, and a working backtesting framework. In particular, the historical, Monte Carlo, and parametric engines were all implemented within a common architecture, allowing the same portfolios to be evaluated under different modeling assumptions. The project also succeeds in extending risk measurement beyond simple stock portfolios by incorporating option repricing, implied-volatility surface construction, realized forward PnL backtesting, and statistical exceedance diagnostics. From a research perspective, one of the most useful outcomes is that the system makes it possible to compare methods directly and to see where they agree, where they diverge, and how their behavior changes across linear, nonlinear, and hedged portfolios.

Several parts of the project worked especially well. The portfolio valuation framework remained consistent across forecasting and backtesting, which made the empirical comparisons more meaningful. The true historical n -day methodology performed well in many of the equity-only backtests and emerged as the most reliable VaR approach overall in that setting. The modular architecture, cache layer, and reporting framework also proved effective, since they made it possible to scale from single calculations to many scenario-based experiments with reproducible saved outputs. The addition of diagnostic plots, exceedance statistics, Kupiec testing, exact binomial tests, and traffic-light classifications also made the results easier to interpret than a single VaR number alone.

At the same time, the project also revealed several limitations. The options-only and mixed portfolio backtests showed that calibration becomes much less stable once nonlinear payoff structure dominates portfolio risk. No single method performed uniformly well across all option portfolios, and approximate neutrality in construction did not always translate into good realized backtest behavior. ES also proved much harder to interpret in a formal backtesting sense than VaR, which is consistent with the theoretical difficulty of testing ES through simple hit-based methods. More broadly, the project showed that results are highly sensitive to volatility treatment, payoff nonlinearity, and exact structure of hedged portfolio.

There are several clear directions for improvement. The most important is richer option-market data, especially longer historical implied-volatility surface histories, which would allow mixed and options-heavy portfolios to be tested across periods such as 2008 and 2021 rather than only in the local 2026 options-data window. The Monte Carlo framework could also be extended to include more realistic volatility dynamics, such as stochastic-volatility or surface-factor models, instead of primarily relying on spot simulation. Likewise, the parametric model could be improved beyond the current local delta-gamma approximation, especially for portfolios with strong curvature or asymmetric payoff structure. On the backtesting side, the framework could be extended with joint VaR/ES procedures or more specialized ES tests, and the effect of overlapping windows could be analyzed more explicitly.

Overall, we can view this project as a successful research-oriented risk engine rather than as a final production system. It demonstrates that realistic portfolio risk analysis for equity and option portfolios can be implemented end-to-end, and it provides a useful platform for comparing methodologies empirically. Its main strengths are transparency, modularity, and the ability to connect model assumptions directly to backtest outcomes. Its main shortcomings are data limitations, simplified volatility modeling in some components, and the difficulty of achieving stable calibration for strongly nonlinear portfolios.